

MISP Web Scraper

CTIS - October 19/20 - 2022

cudeso.be
We Secure You

<https://www.cudeso.be>
koen.vanimpe@cudeso.be



- **Freelancer**

- Incident response, threat intelligence, security monitoring

- **Open source contributions**

- MISP modules, taxonomies, automation and integration with DFIR tools, ...
- “*MISP tip-of-the-week*”

- **BelgoMISP**

- Belgian MISP User Group

- **OSINT threat feed**

- botvrij.eu

✉ koen.vanimpe@cudeso.be

🔗 <https://www.cudeso.be>

🔗 <https://www.vanimpe.eu>

🐙 <https://github.com/cudeso>

🐦 @cudeso

Why?

How it started: adding threat events to botvrij.eu

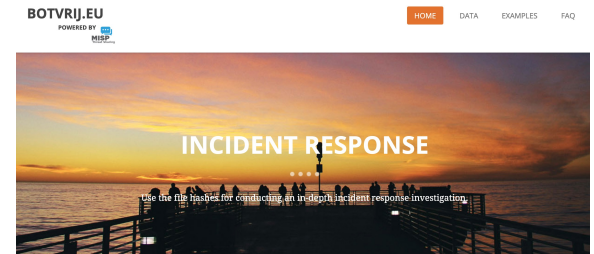
Public reports

Blog posts

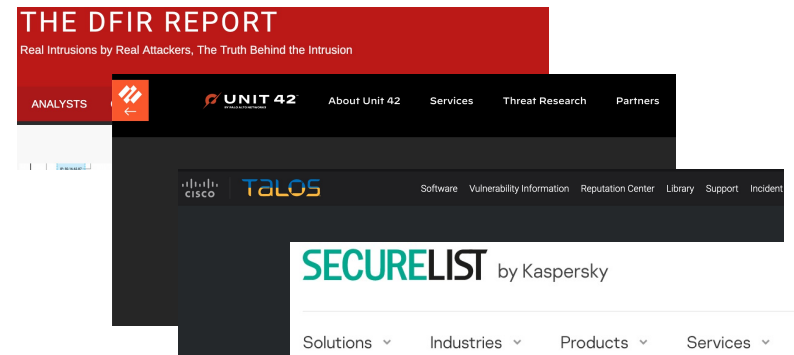
Whitepapers

“OSINT”

 osint:source-type="blog-post" x  tlp:white x  type:OSINT x

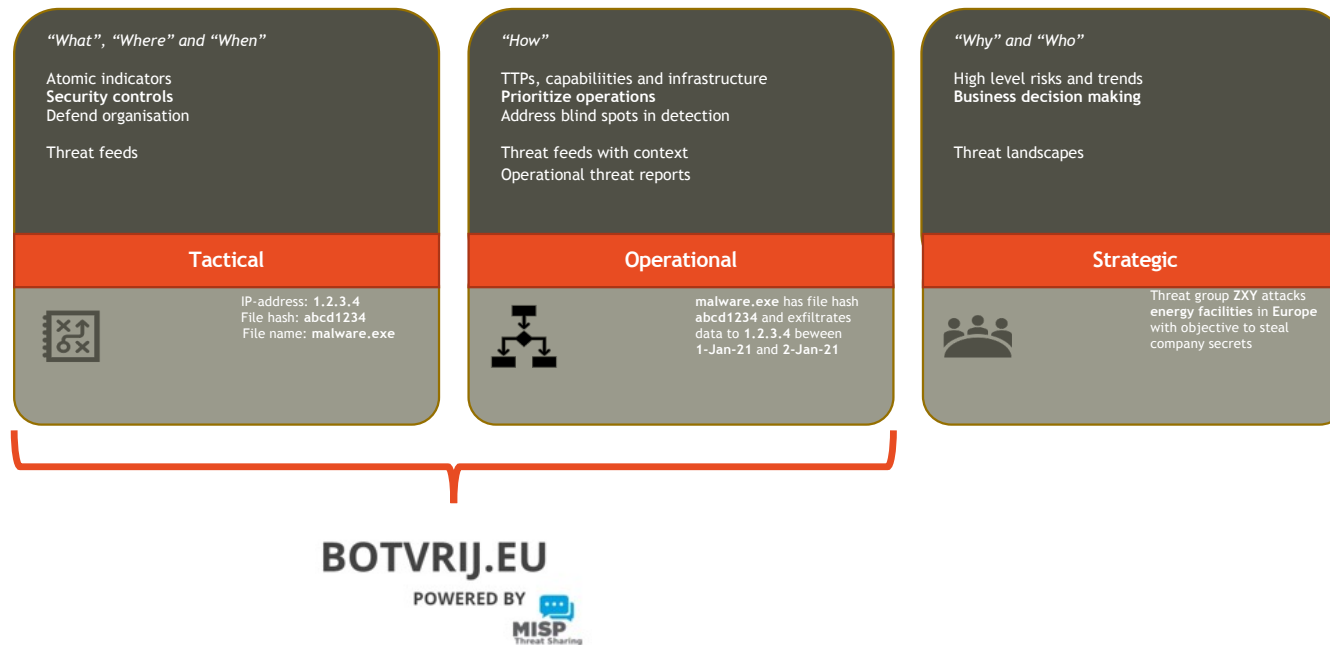


- A lot of **copy and pasting**
 - Tedious
 - Time consuming
 - Risk of errors



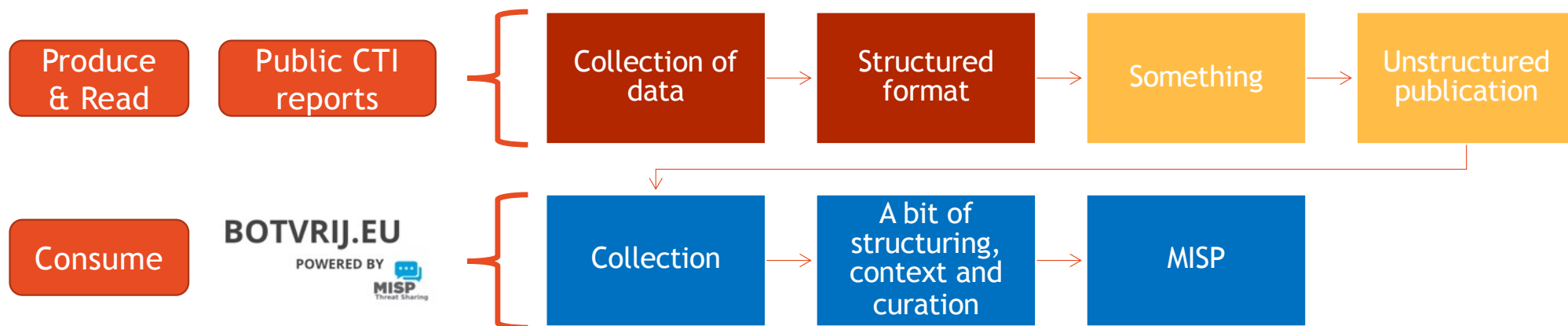
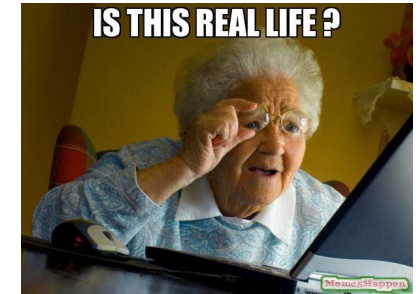
Avoid being just an indicator list

- Focus on **tactical** and **operational** intelligence
- Include context for events and attributes



Sources used for the OSINT feed

- A lot of public reports *mostly* contain **unstructured** information
 - Or structured but not consistent in format
 - CSV in format x, CSV in format y, TXT files
- Reports sometimes **focus more on readers**, than on consumers
- Workflow (presumably not just for me ...)



Why are public sources so difficult to consume?

- It's not about *solving the sharing problem*
 - But about making public reports more easily consumable
- There are commercial interests
 - But providing structured information publicly can be a valuable “**teaser**” for your professional CTI services
- Not all is bad



A screenshot showing three GitHub repository pages side-by-side. The first is 'eset/malware-ioc' (Public), showing a commit by 'porolli and eset-research' adding IoCs for POLONIUM, with a list of folders like 'amavaldo', 'animalfarm', 'attor', and 'backdoordiplomacy'. The second is 'pan-unit42/iocs' (Public), showing a commit by 'chkroot' updating Qakbot, with a list of folders like 'AcidBox', 'ArtraDownloader', and 'BazarCall'. The third is 'NVISOsecurity/nviso-cti' (Public), showing a commit by 'sylokttools' adding IoCs for September's Generic Threat Briefing, with a list of files like 'Iranian_Threat_Actor_IOCs.csv' and 'bumblebeeRoundTwo.csv'.

What would make this work easier?

- **Formats?**
 - **STIX**
 - Low adoption
 - Certainly what concerns public reports
 - **CSV/TXT**
 - Not always consistent
 - Field sequence, naming conventions
 - **PDF/XLS/...**
 - No



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 - No
- Pending everyone starts using the **MISP core format** ...



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 - Certainly what concerns put

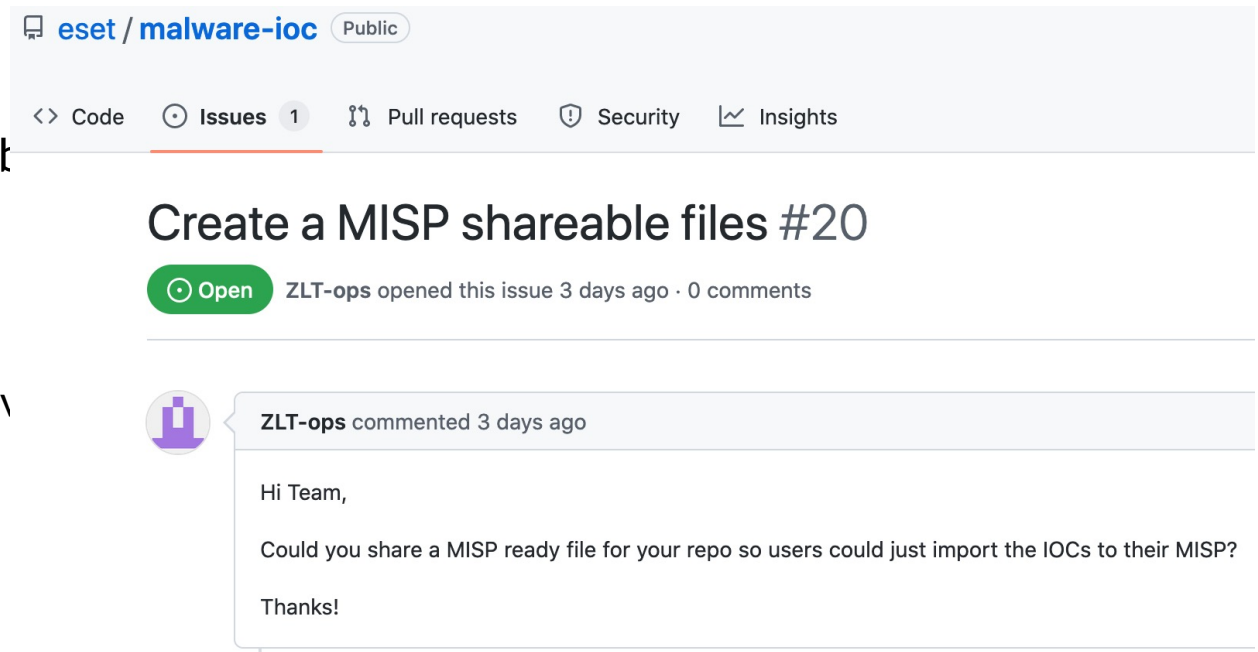
- **CSV/TXT**

- Not always consistent
 - Field sequence, naming conv

- **PDF/XLS/...**

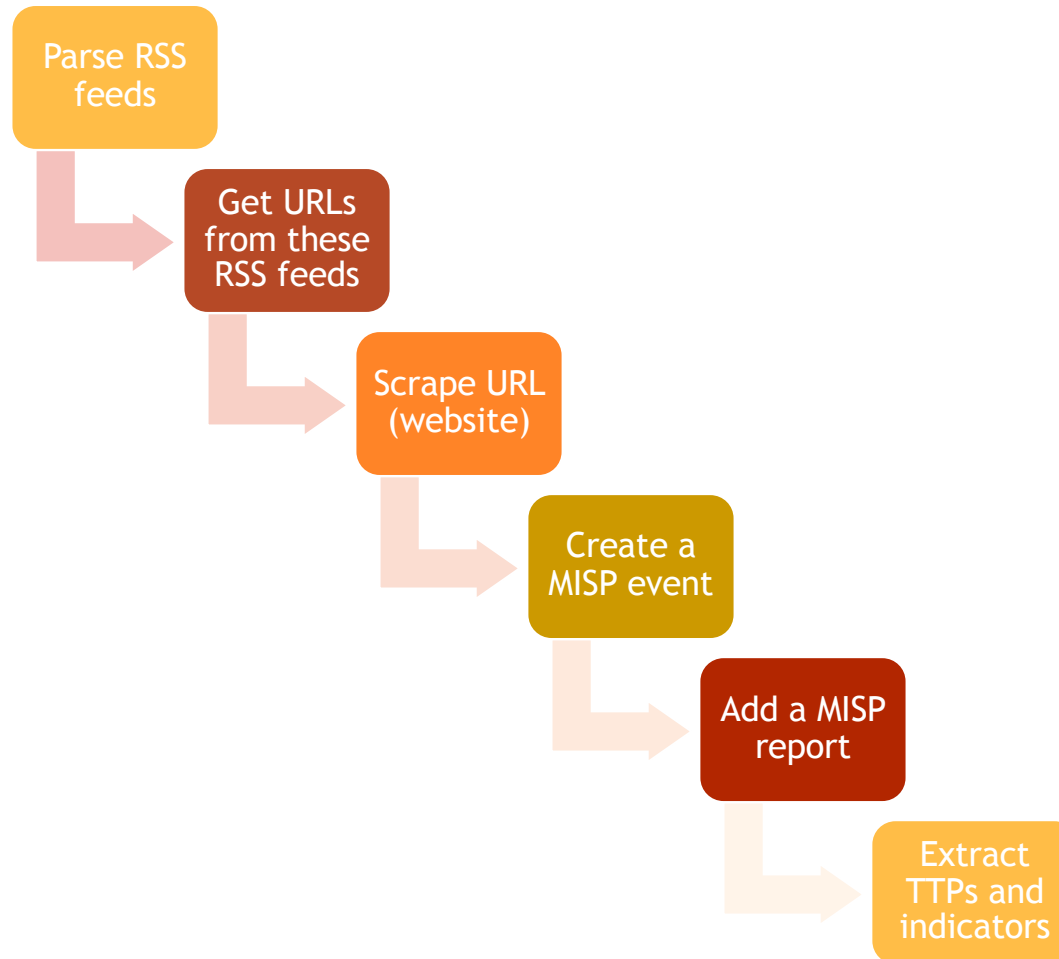
- No

- Pending everyone starts using the **MISP core format ...**

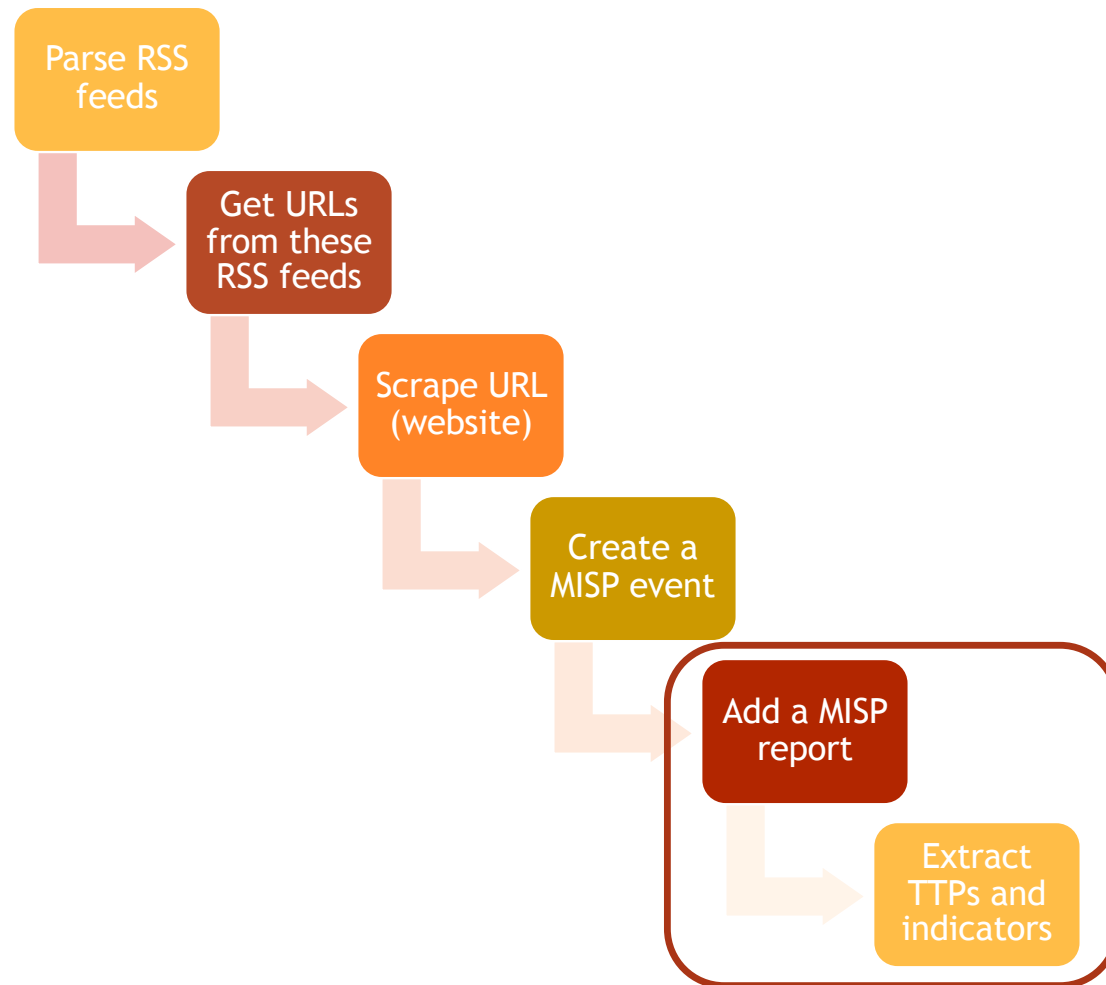


MISP web scraper

MISP web scraper



MISP web scraper



Build on MISP reports

"CTI report"

References to
attributes and
objects

References to
taxonomies and
galaxy matrices

In raw text or
markdown

Is part of an
event and
distributed with
events

Build on MISP reports

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1

Indicators

TTPs

Build on MISP reports

"CTI report"

References to
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events

1

Indicators

TTPs

2

Convert "website" content to markdown

main ▾

[misp-modules](#) / [misp_modules](#) / [modules](#) / [expansion](#) / [html_to_markdown.py](#)

Event Reports

Import from URL

Build on MISP reports

"CTI report"

References to
attributes and
objects

References to
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In raw text or
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Indicators

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Convert "website" content to markdown

main ▾

[misp-modules](#) / [misp_modules](#) / [modules](#) / [expansion](#) / [html_to_markdown.py](#)

Event Reports

Import from URL

3

Save

Menu ▾

Help

Download

Markdown parsing rules

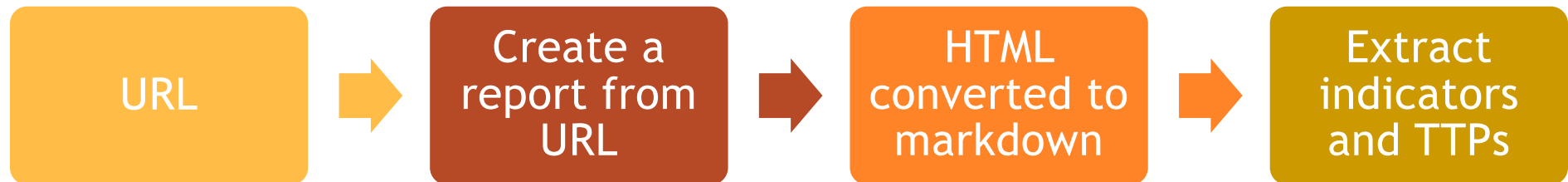
Markdown rendering rules

Extract entities

Manual extraction

Automatic extraction

Easy wins with the MISP report feature



MISP web scraper components

Flask

- Web form
- **Manual:** add a URL to Redis

MISP-Scraper

Go to MISP

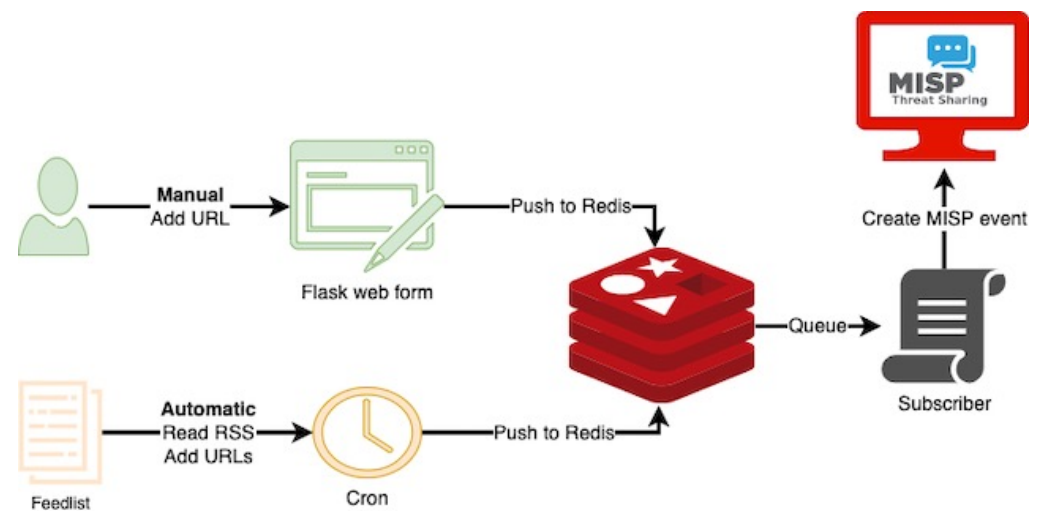
Add either a page title and URL or raw HTML to submit to MISP Scraper

Feed title
Feed URL

Page title

Page URL (link)
(not scraped when raw HTML is submitted)

Raw HTML
(ignored when empty, then Page URL is scraped)



MISP web scraper components

Flask

- Web form
- **Manual:** add a URL to Redis

Cron job

- Refresh RSS feed list (*feedlist.json*)
- **Automatic:** add URLs to Redis

Redis

- Push URLs to Redis

Subscribe to Redis queue

- Pick up URLs, crawl the website
- **Create MISP event and report**

MISP-Scraper

Go to MISP

Add either a page title and URL or raw HTML to submit to MISP Scraper

Feed title

Feed URL

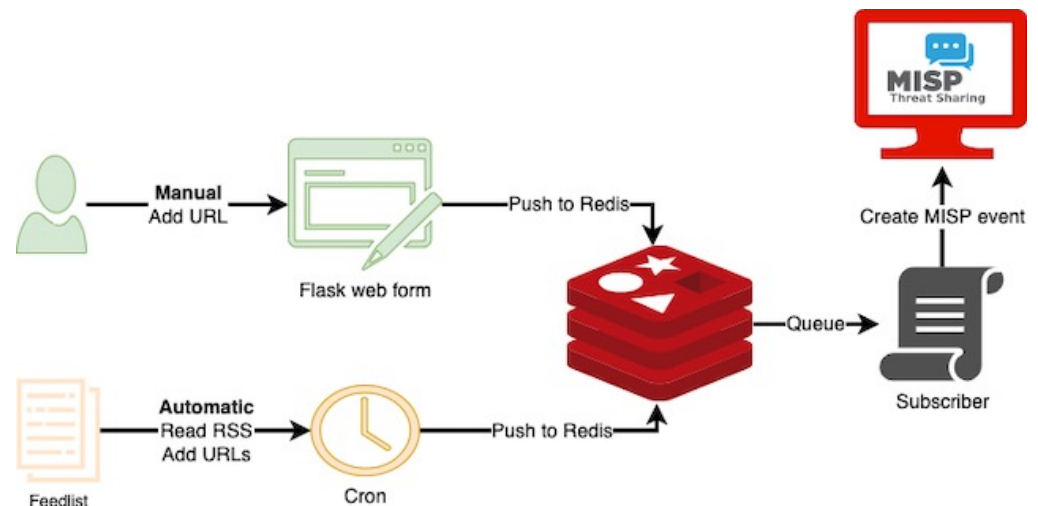
Page title

Page URL (link)

(not scraped when raw HTML is submitted)

Raw HTML

(ignored when empty, then Page URL is scraped)

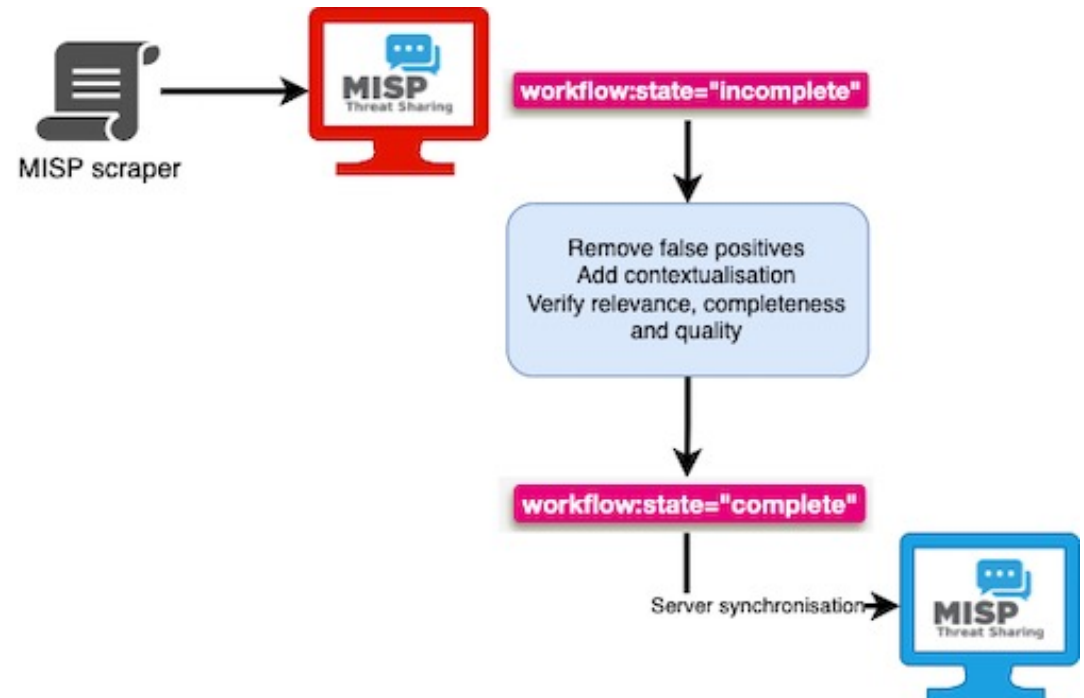


Web scraper workflow

- **Scrape** in a “dirty” environment

- Automated, “cron job”
- Create MISP event

 workflow:state="incomplete" 



Web scraper workflow

- **Scrape** in a “dirty” environment

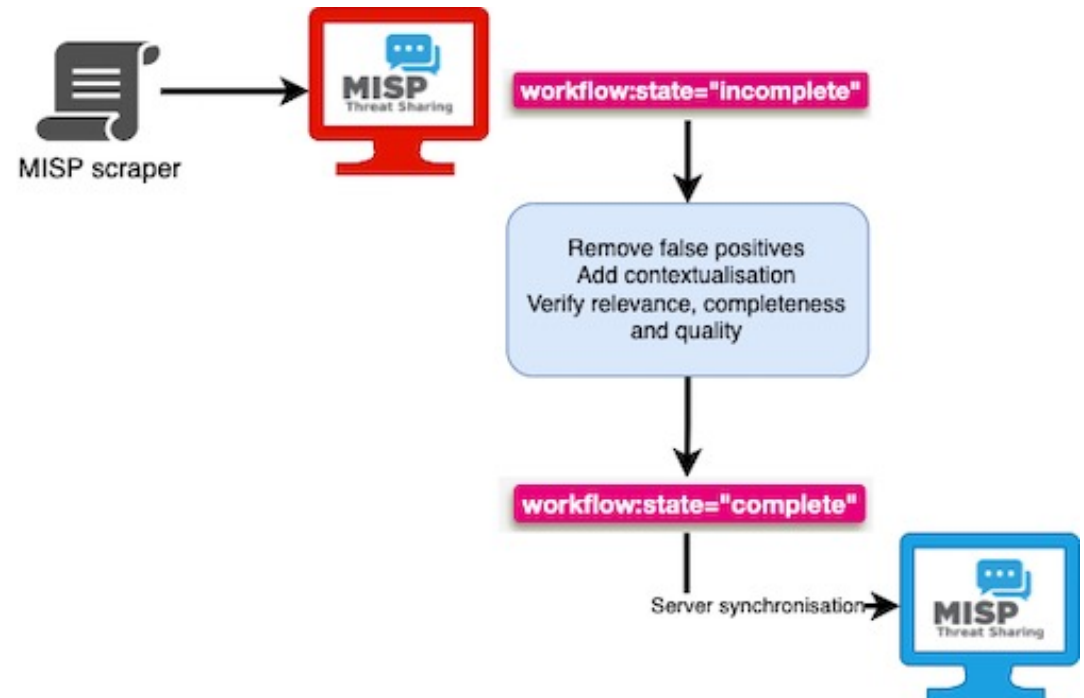
- Automated, “cron job”
- Create MISP event

 workflow:state="incomplete" 

- **Curate** event

- Scrape successful?
- Relevant and complete?
- Remove false positives

 workflow:state="complete" 



Web scraper workflow

- **Scrape** in a “dirty” environment

- Automated, “cron job”
- Create MISP event

 workflow:state="incomplete" 

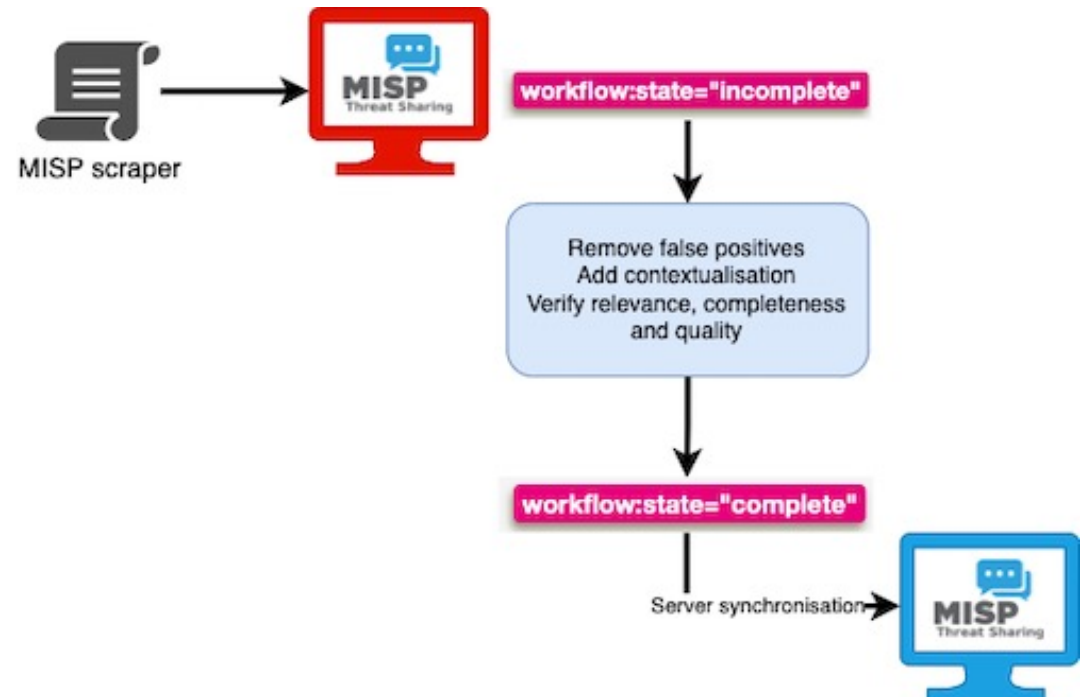
- **Curate** event

- Scrape successful?
- Relevant and complete?
- Remove false positives

 workflow:state="complete" 

- **Publish and synchronise**

- With your “clean” MISP



Curate scraped events (1/2)

- Analyst checks
 - Scrape successful?
 - HTTP error codes added as tags
 - Not “blocked” by a CDN

misp-scrafer:HTTP=403

misp-scrafer:HTTP=404

html .st0{display:none}.st1{fill:#006db6} # Access denied

You cannot access **hostname**. Refresh the page or contact the site owner to request access.

- Ray ID: **ip-dst**
- Timestamp: 2022-10-12 08:30:40 UTC
- Your IP address: **ip-dst**
- Requested URL: @[suggestion](**hostname**) /2022/10/threat-roundup-0930-1007.html)
- Error reference number: 1020
- mitre-attack-pattern** → **Server - T1583.004** ID: **regkey FL_78F89**
- User-Agent: **url python-requests/2.27.1**
- Cloudflare Ray ID: **ip-dst** Your IP: **ip-dst**

- Relevant?
 - “Threat report”?

☐	Date ↑	Org	Category	Type	Value
☐	2022-10-12		Other	comment	Preventing Cryptocurrency Cyber Extortion
☐	2022-10-12		External analysis	link	http://feeds.trendmicro.com/TrendMicroSimplySecurity
☐	2022-10-12		External analysis	link	https://www.trendmicro.com/en_us/ciso/22/i/prevent-cyber-extortion.html ⚠

Page 1 of 1, showing 1 records out of 3 total, starting on record 1, ending on 3

- Wrong attribute category or types

Payload delivery hostname 2022-09-23-iocs-for-icedid-and-cobalt-strike.txt.zip

Curate scraped events (2/2)

- **False positives**
 - MISP warninglists
- **Mass edit attributes**
 - Remove to_ids
 - Attributes primarily fit for context, not for detection or filtering
 - Remove correlation
- **Mass delete attributes**
 - Attributes irrelevant for the threat event

Mass Edit Attributes

Distribution
Do not alter current settings ▾

For Intrusion Detection System
No ▾

☐ Create proposals

Correlations
Disable correlations ▾

Payload delivery	filename	Trojan.BAT.ADFIND.YECGUT
Payload delivery	filename	Backdoor.Win32.SYSTEMBC.YXCFLZ
Payload delivery	filename	Backdoor.Win32.COBEACON.YXCH3
Payload delivery	filename	Ransom.Win32.PLAYCRYPT.YECGUT
Payload delivery	filename	Trojan.Win64.PRIVICMD.YXCHW
Payload delivery	filename	Ransom.Win32.PLAYDE.YACHWT 

Make curation a bit easier (1)

- **Doubles**

- Web scraper: **skip MISP events** already created
 - (based on event title + scraper tag)

- **Retention**

- Old events are automatically deleted
- Avoid analyst overload

retention:2d

retention:7d

- **Classification of scraped events**

 **misp:tool="misp-scraper"**   **osint:source-type="blog-post"**   **retention:2d** 
 **misp:event-type="collection"**   **tlp:white**   **workflow:state="incomplete"**   **scraper:TrendMicro** 

- Tool and source
 - Unique tag per source
 - Allows f.e. to filter for all posts scraped from the source TrendMicro
- Default TLP and workflow state

Make curation a bit easier (2)

- Custom **warninglist**
 - Via a MISP warninglist
 - Avoid adding known non-relevant attributes
 - URLs to partner websites
 - Social media
 - “Commercial” strings
- Links and comment attributes to describe source
 - Feed, title
 - No correlation
 - Link to web page
 - Correlation

SCRAPER

ID	72
Name	Scraper
Description	Warninglist to be used for the web scraper
Version	9
Category	False positive
Type	string
Accepted attribute types	
Enabled	Yes Disable

Values

<http://club.orbisius.com/products/wordpress-plugins/whitelist-ip-for-limit-lc>
<https://t.co/VVpmpTPS2T>
<https://threatpost.com/malware-traffic-analysis.net>
payloads.This
Zone.Identifier
src="/static/blogv2/js/vendors.js
/static/blogv2/js/vendors.js
<http://google.com/ads/remarketingsetup>
threatpost.com

External analysis	link	https://securelist.com/feed/ ⚠	   	Feed URL	☐	
External analysis	link	https://securelist.com/malicious-whatsapp-mod-distributed-through-legitimate-apps/107690/ ⚠	   	Blog URL	☒	
Other	comment	Malicious WhatsApp mod distributed through legitimate apps	   	Blog title	☐	

Make curation a bit easier (2)

- Custom warninglist
 - Via a MISP warninglist
 - Avoid adding known non-relevant attributes
 - URLs to partner websites
 - Social media
 - “Commercial” strings

Curation: events with less than 4 attributes can be ignored

- Links and comment attributes to describe source report
 - Feed, title
 - No correlation
 - Link to web page
 - Correlation

SCRAPER

ID	72
Name	Scraper
Description	Warninglist to be used for the web scraper
Version	9
Category	False positive
Type	string
Accepted attribute types	
Enabled	Yes Disable

Values

<http://club.orbisius.com/products/wordpress-plugins/whitelist-ip-for-limit-lc>
<https://t.co/VVpmpTPS2T>
<https://threatpost.com/malware-traffic-analysis.net>
payloads.This
Zone.Identifier
src="/static/blogv2/js/vendors.js
/static/blogv2/js/vendors.js
<http://google.com/ads/remarketingsetup>
threatpost.com

External analysis	link	https://securelist.com/feed/ ⚠			Feed URL	☐	
External analysis	link	https://securelist.com/malicious-whatsapp-mod-distributed-through-legitimate-apps/107690/ ⚠			Blog URL	☒	
Other	comment	Malicious WhatsApp mod distributed through legitimate apps			Blog title	☐	

Scraped event

Preventing Black Basta Ransomware in 2022

Event ID	294
UUID	391357f3-d319-41f8-8525-ce2b083097b6  
Creator org	CUDESO
Owner org	CUDESO
Creator user	
Protected Event (experimental) 	Tags from scraper to ease curation
Tags	 misp:tool="misp-scraper"   osint:source-type="blog-post"   misp:event-type="collection"   misp-galaxy:botnet="Qbot"   misp-galaxy:malpedia="Black Basta"   workflow:state="complete"   tlp:white     
Date	2022-09-12

Scraped event

Preventing Black Basta Ransomware in 2022

Event ID

294

UUID

391357f3-d319-41f8-8525-ce2b083097b6

Creator org

CUDES0

Owner org

CUDES0

Creator user

Protected Event
(experimental)

Tags

misp:tool="misp-scra

per" x

osint:source-type="blog-post" x

misp:event-type="collection" x

misp-galaxy:botnet="Qbot" x

misp-galaxy:malpedia="Black Basta" x

workflow:state="complete" x

tlp:white x

Date

2022-09-12

Event Reports

+ Add Event Report

Import from URL

Generate report from Event

All

Default

Deleted

ID	Name	Last update	Distribution	Actions
16	Report from - https://www.deepinstinct.com/blog/black-basta-ransomware-threat-emergence (1662992420)	2022-09-12 16:20:20	Inherit event	

Tags from scraper to ease curation

Extracted attributes

☐	2022-09-12	Network activity	url	https://aazsbsgya565vlu2c6bzy6yflebkcbtvvctvolt33s77xypl7nypxyd.onion/		Black Basta TOR site	☒	☒	Inherit
☐	2022-09-12	Payload delivery	sha256	9fce9ee85516533bae34fc1184a7cf31fa9f2c7889b13774f83d1df561708833		Black Basta Windows Variant	☒	☒	Inherit
☐	2022-09-12	Payload delivery	sha256	203d2807df6ef531efbec7bfd109986de3e23df64c01ea4e337cbe5ba675248b		Black Basta Windows Variant	☒	☒	Inherit
☐	2022-09-12	Payload delivery	sha256	0d6c3de5aebb8e85939d7588150edf7b7bdc712fceb6a83d79e65b6f79bfc2ef		Black Basta Linux/ESXi Variant	☒	☒	Inherit
☐	2022-09-12	Payload delivery	sha1	2bee3f716b80273db9639376a296cf19cdba0f1a		Qbot payload certificate thumbprint (sha1)	☒	☒	Inherit
☐	2022-09-12	Network activity	url	http://146.70.79.52/		Payload host	☒	☒	Inherit

Scraped event

Ragnar Locker Ransomware Targeting the Energy Sector

Event ID	292
UUID	8dbaaaac-a671-4a02-8dab-5eec2a1c935b  
Creator org	CUDESO
Owner org	CUDESO
Creator user	koen.vanimpe@cudeso.be
Protected Event (experimental) 	 Event is in unprotected mode.
Tags	 <code>misp:tool="misp-scraper"</code>   <code>osint:source-type="blog-post"</code>   <code>misp:event-type="collection"</code>   <code>misp-galaxy:malpedia="RagnarLocker (Windows)"</code>   <code>workflow:state="complete"</code>   <code>tlp:white</code>   
Date	2022-09-12

Scraped event

Ragnar Locker Ransomware Targeting the Energy Sector

Event ID 292

UUID 8dbeaac-a671-4a02-8dab-5eec2a1c935b 

Creator org CUDES0

Owner org CUDES0

Creator user koen.vanimpe@cudeso.be

Protected Event  Event is in unprotected mode.
(experimental) 

Tags  **misp:tool="misp-scraper"**   **osint:source-type="blog-post"**   **misp:event-type="collection"** 
 **misp:galaxy:malpedia="RagnarLocker (Windows)"**   **workflow:state="complete"**   **tlp:white** 
 

Date 2022-09-12

Event Reports

[+ Add Event Report](#) [Import from URL](#) [Generate report from Event](#) [All](#) [Default](#) [Deleted](#)

ID	Name	Last update	Distribution	Actions
14	Report from - https://www.cybereason.com/blog/threat-analysis-report-ragnar-locker-ransomware-targeting-the-energy-sector (1662984886)	2022-09-12 14:14:47	Inherit event	 

☐	2022-09-12	Payload delivery	filename	ntuser.dat.log	 	 	☒	☐
☐	2022-09-12	Payload delivery	filename	bootfront.bin	 	 	☒	☐
☐	2022-09-12	Payload delivery	sha256	9b62cdb5714c34924333dfa3baefd993efeb68109580b682b074f0e73b63983	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	c2bd70495630ed8279de0713a010e5e5f3da29323b59ef71401b12942ba5216	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	5469182495d92a5718e0e1dcd371e92b79724e4270501541318de693d341c89	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	ec35c76ad2c8192f09c02eca1f263b406163470ca8438d054db7adcf5bfc0597	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	dd5d4cf9422b6e4514d49a3ec542cfeb682be8a24079010cda689afbb44ac0f4	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	cf5ec678a2f836f859eb983eb633d529c25771b3b7505e74aa695b7ca00f9fa8	 	  Ragnar Locker Binary	☒	☒
☐	2022-09-12	Payload delivery	sha256	ce33096639fb5c51684e9e3a7c7c7161884ecad29e8d6ad602fd8be42076b8d4	 	  Ragnar Locker Binary	☒	☒

What works well?

Always

- Hashes
- IP address
- Easy to identify with regexp

Often

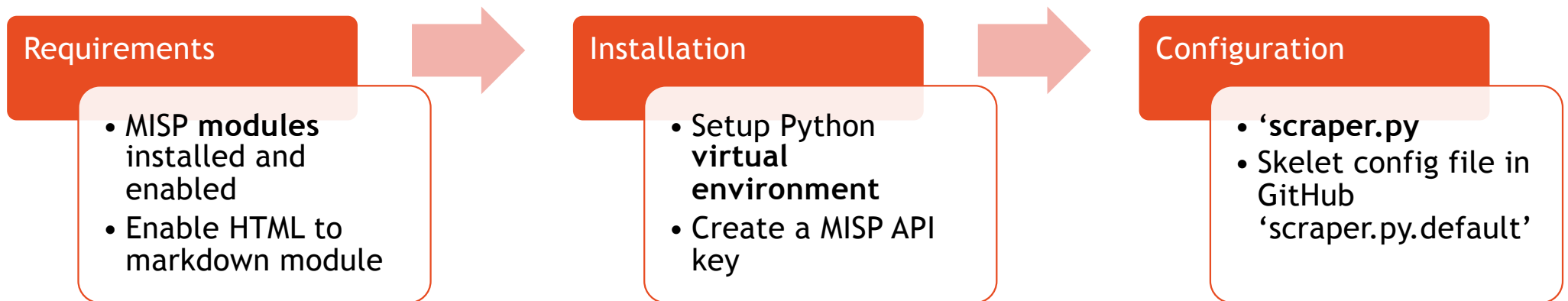
- URLs
- Domain and hostname
- E-mail src
- TTPs
 - If not part of the commercial text on web pages

Sometimes

- Filenames
- Registry keys

- Good for indicator collection
- Some automatic TTPs
- Converting to objects remains a manual process
- Detection rules (yara) require manual edition
- **RSS reader for threat reports**

Installation and configuration



```
git clone https://github.com/cudeso/misp-scraper
cd misp-scraper
virtualenv scraper
source scraper/bin/activate
pip install -r requirements.txt
cp scraper.py.default scraper.py
```

Configuration file

General configuration

```
misp_scraper_log = "/var/log/misp/scraper.log" # Log file for the scraper
app_name = "MISP-Scraper" # Used in the Flask server
manual_feed = "Manual"
manual_feedsources = "Manual"
flask_secret_key = "MyZsuePerSecret1984key" # Flask secret key. Change this to
flask_address = "127.0.0.1" # IP address for the Flask server
flask_port = 5200
flask_certificate_file = "/etc/ssl/private/misp.local.crt" # Point to a certificate file (fe.
flask_certificate_keyfile = "/etc/ssl/private/misp.local.key" # Point to a certificate key file
logging_level = "error"
feedlist = "/home/ubuntu/misp-scraper/feedlist.json" # Location of the RSS feedlist
rawhtml_distribution = 4
rawhtml_sharing_group_id = 4
```

MISP configuration

```
misp_key = "" # MISP API key
misp_url = "" # MISP url
misp_verifycert = False
misp_distribution = 2 # Connected communities
misp_threat_level_id = 2 # Medium
misp_analysis_level = 2 # Completed
misp_scraper_event = "Scraper" # Prefix of the MISP event
misp_retentiontime = "99d" # Set this to >days than URLs are
misp_scraper_tags = ["misp:tool=\"misp-scraper\"", "osint:source-type=\"blog-post\"", "misp:event-
misp_warninglist = 0
misp_hard_delete_on_cleanup = False
```

Feedlist configuration file

JSON

List of feeds

- title
 - Used in the tagging of events
- url
 - URL to download the RSS feed

```
{  
  "feeds": [  
    {  
      "title": "Proofpoint",  
      "url": "https://www.proofpoint.com/us/threat-insight-blog.xml"  
    },  
    {  
      "title": "Red Canary",  
      "url": "https://www.redcanary.co/feed/"  
    },  
    {  
      "title": "vanimpe.eu",  
      "url": "https://www.vanimpe.eu/feed/rss/"  
    },  
    {  
      "title": "Threatpost",  
      "url": "http://threatpost.com/feed"  
    }  
  ]  
}
```

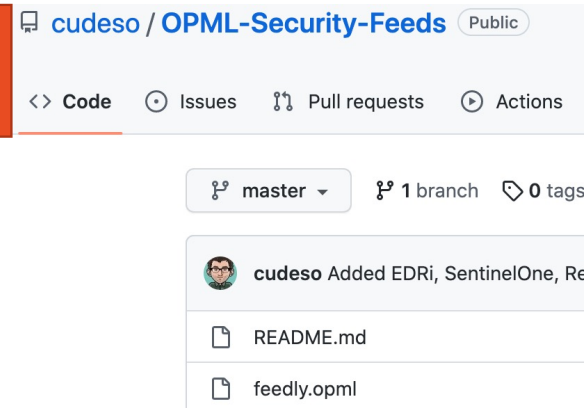
Feedlist configuration file

JSON

List of feeds

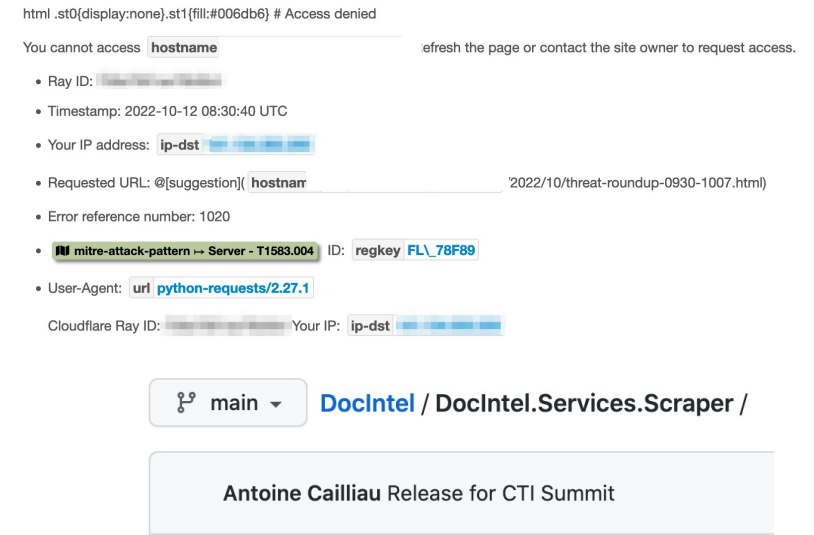
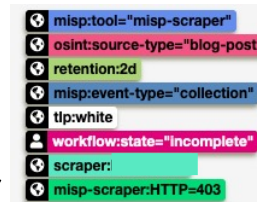
- title
 - Used in the tagging of events
- url
 - URL to download the RSS feed

- Need inspiration for RSS feeds?
 - <https://github.com/cudeso/OPML-Security-Feeds>



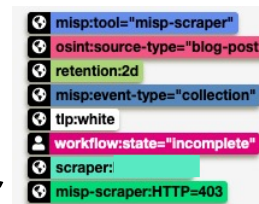
Room for improvement (1)

- Web crawler is **Python**
- User-agent blocked
- Improve with **ail-splash-manager**
 - Instead of handling the crawling in web-scraper, have AIL do the crawling
 - Microsoft Playwright ^^AIL



Room for improvement (1)

- Web crawler is **Python**
 - User-agent blocked
 - Improve with **ail-splash-manager**
 - Instead of handling the crawling in web-scraper, have AIL do the crawling
 - Microsoft Playwright ^^AIL
 - First steps done by allowing to submit raw HTML



html.st0(display:none).st1(fill:#006db6) # Access denied

You cannot access `hostname`. Please refresh the page or contact the site owner to request access.

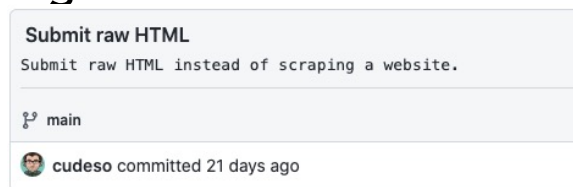
- Ray ID: `ip-dst`
- Timestamp: 2022-10-12 08:30:40 UTC
- Your IP address: `ip-dst`
- Requested URL: `@[suggestion](hostname)2022/10/threat-roundup-0930-1007.html`
- Error reference number: 1020
- `mitre-attack-pattern -> Server - T1583.004` ID: `regkey FL_78F89`
- User-Agent: `url python-requests/2.27.1`

Cloudflare Ray ID: `ip-dst` Your IP: `ip-dst`

[main](#) [DocIntel](#) / DocIntel.Services.Scraper /

Antoine Cailliau Release for CTI Summit

- Include **screenshots**
 - Via ail-splash-manager?



- Deal with RSS sources containing ridiculous number of URLs
 - Feeds containing all articles going back to 20xx
 - Discover publish date and only consider recent reports

Room for improvement (2)

- Use Natural Language Processing (NLP) to analyse reports
 - FIRSTCON 2022

All the Unstructured Data! Using NLP to Process Threat Reports

TLP:AMBER  ^

Patrick Grau (Bosch, DE)

- **Bulk change category and type** of attributes
 - Similar as to bulk edit events
 - Would in general be a great feature to edit MISP events
- Use Feedly AI
 - Remove non-relevant clutter text from web pages

Room for improvement (3)

- Automatically **remove attributes** for specific scraped sources
 - F.e. always remove attribute xyz from source abc
 - Company labels etc.
- Remove specific **CSS classes**

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- Automatically **remove attributes** for specific scraped sources
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- Remove specific **CSS classes**
 - Need galaxies/clusters?

Misinformation Pattern Q

Instagram Q:   
Facebook Q:   
Twitter Q:   

Target Information Q

Australia Q:   
Canada Q:   
France Q:   
Hong Kong Q:   
Indonesia Q:   
Ireland Q:   
Malaysia Q:   
New Zealand Q:   
Philippines Q:   
Singapore Q:   
South Africa Q:   
United Kingdom Q:   
United States Q:   

Sector Q

Development Q:   
Education Q:   
Electric Q:   
Gas Q:   
Health Q:   
Industrial Q:   
Insurance Q:   
Intelligence Q:   
Legal Q:   
Manufacturing Q:   
Oil Q:   
Technology Q:   
Water Q:   

Room for improvement (3)

- Automatically remove attributes for specific scraped sources
 - F.e. always remove attribute xyz from source abc
 - Company labels etc.
- Remove specific CSS classes
 - Need galaxies/clusters?
 - Mass delete clusters

```
<div class="social-copyright-row">  
  <div class="inner-container">  
    <div class="row">  
      <ul class="col-md-6 social-media-links">
```

```
<div class="dropdown stretched-dropdown">  
  <button class="menu-button button-default" type="button" data-toggle="dropdown">  
    <span class="menu-button__icon icon-region"></span>  
    <span class="menu-button__text">Region</span>  
  </button>  
  <div class="dropdown-menu align-left">  
    <ul class="menu-column col-xs-12 col-sm-4 col-md-3">  
      <li class="dropdown-header">  
        The Americas  
      </li>  
      <li>  
        <a href="/en_us.html">  
          United States  
        </a>  
      </li>  
      <li>  
        <a href="/pt_br.html">  
          Brasil  
        </a>  
      </li>  
    </ul>  
  </div>  
</div>
```

Misinformation Pattern Q

Instagram Q
Facebook Q
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Target Information Q

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Sector Q

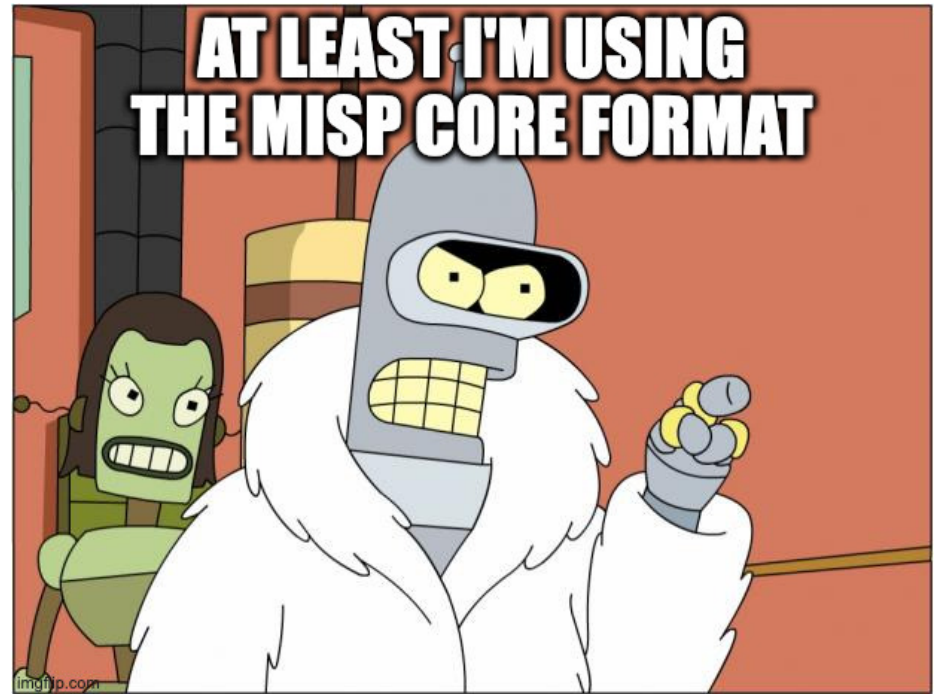
Development Q
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Health Q
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Water Q

Room for improvement (4)

- Scraping websites to gather threat intel reports is far from ideal.

Room for improvement (4)

- Scraping websites to gather threat intel reports is far from ideal.
 - MISP web scraper is no longer needed and everyone uses the MISP core format.



Get it via GitHub

- <https://github.com/cudeso/misp-scrapers>

Documentation on GitHub and on the MISP project website

- <https://www.misp-project.org/2022/08/08/MISP-scrapers.html>

Questions?